DEWAGEDARA D.M.E.S.
E/21/087
LAB 08

TASK 01

A.

```
mysql> DESCRIBE employees;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_no     | int           | NO   | PRI | NULL    |       |
| birth_date | date          | YES  |     | NULL    |       |
| first_name | varchar(14)   | YES  |     | NULL    |       |
| last_name  | varchar(16)   | YES  |     | NULL    |       |
| sex        | enum('M','F') | YES  |     | NULL    |       |
| hire_date  | date          | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
6 rows in set (0.02 sec)
```

B.

```
mysql> SHOW INDEX FROM employees;
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table     | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| employees |          0 | PRIMARY  |            1 | emp_no      | A         |      299433 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
1 row in set (0.04 sec)
```

C.

- The DESCRIBE command shows the table structure, data types, and constraints.
- SHOW INDEX reveals what indexes currently exist (only PRIMARY KEY on emp_no).
- Most columns don't have indexes initially, which affects query performance.

D.

```
mysql> SELECT first_name FROM employees ORDER BY first_name;

+----------------+
300024 rows in set (0.32 sec)
```

E.

```
mysql> EXPLAIN SELECT first_name FROM employees ORDER BY first_name;
+----+-------------+-----------+------------+------+---------------+------+---------+------+--------+----------+----------------+
| id | select_type | table     | partitions | type | possible_keys | key  | key_len | ref  | rows   | filtered | Extra          |
+----+-------------+-----------+------------+------+---------------+------+---------+------+--------+----------+----------------+
|  1 | SIMPLE      | employees | NULL       | ALL  | NULL          | NULL | NULL    | NULL | 299433 |   100.00 | Using filesort |
+----+-------------+-----------+------------+------+---------------+------+---------+------+--------+----------+----------------+
1 row in set, 1 warning (0.01 sec)
```

F.

- The EXPLAIN output shows how MySQL executes the query.
- Without an index on first_name, MySQL must:
    1. Scan the entire table (type: ALL)
    2. Sort all results (Using filesort)
    3. This is inefficient for large tables

G.

```
mysql> CREATE INDEX fname_index ON employees(first_name);
Query OK, 0 rows affected (1.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

H.

```
| Zvonko          |
| Zvonko          |
+-----------------+
300024 rows in set (0.19 sec)
```

I.

```
mysql> EXPLAIN SELECT first_name FROM employees ORDER BY first_name;
+----+-------------+-----------+------------+-------+---------------+-------------+---------+------+--------+----------+-------------+
| id | select_type | table     | partitions | type  | possible_keys | key         | key_len | ref  | rows   | filtered | Extra       |
+----+-------------+-----------+------------+-------+---------------+-------------+---------+------+--------+----------+-------------+
|  1 | SIMPLE      | employees | NULL       | index | NULL          | fname_index | 59      | NULL | 299433 |   100.00 | Using index |
+----+-------------+-----------+------------+-------+---------------+-------------+---------+------+--------+----------+-------------+
1 row in set, 1 warning (0.00 sec)
```

J.

- The query now uses the index (type: index)
- No filesort needed since data is already sorted in the index
- Significant performance improvement

K.

- Before Index: Full table scan + filesort
- After Index: Direct index access, pre-sorted data
- Improvement: Faster execution, especially noticeable with large datasets

Task 02

A composite (multi-column) index is an index that includes multiple columns from a table, allowing efficient queries that filter or sort by multiple columns simultaneously.

A.

Difference from Multiple Single-Column Indexes,

1. Composite Index: Single index structure covering multiple columns
2. Multiple Single Indexes: Separate index structures for each column
3. Advantage: Composite indexes can satisfy queries involving multiple columns in one lookup

B.

Column Order Impact,

1. Left-most prefix rule: Index can only be used if query includes leftmost columns

   Example: Index on (last_name, first_name) can help:

   WHERE last_name = 'Smith'

   WHERE last_name = 'Smith' AND first_name = 'John'

   But NOT: WHERE first_name = 'John' (without last_name)

C.

- Queries frequently filter by multiple columns together
- Sorting by multiple columns
- When query selectivity improves with multiple columns
- Covering indexes (index contains all needed columns)

TASK 03

A.

```
mysql> SELECT * FROM departments WHERE dept_name = 'Finance';
+---------+-----------+
| dept_no | dept_name |
+---------+-----------+
| d002    | Finance   |
+---------+-----------+
1 row in set (0.01 sec)
```

B.

```
mysql> SELECT * FROM departments WHERE dept_no = 'd002';
+---------+-----------+
| dept_no | dept_name |
+---------+-----------+
| d002    | Finance   |
+---------+-----------+
1 row in set (0.00 sec)
```

Conclusions,

- If dept_no is PRIMARY KEY: Very fast lookup (type: const)
- If dept_name has no index: Full table scan (type: ALL)
- Primary keys provide the fastest access method
- Non-indexed columns require full table scans

TASK 04

A.

1.

```
mysql> DESCRIBE titles;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| emp_no    | int          | NO   | PRI | NULL    |       |
| title     | varchar(50)  | NO   | PRI | NULL    |       |
| from_date | date         | YES  |     | NULL    |       |
| to_date   | date         | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

```
mysql> SHOW INDEX FROM titles;
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
| Table  | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
| titles |          0 | PRIMARY  |            1 | title       | A         |           5 |     NULL |   NULL |      | BTREE      |         |              | YES     | NULL       |
| titles |          0 | PRIMARY  |            2 | emp_no      | A         |      426987 |     NULL |   NULL |      | BTREE      |         |              | YES     | NULL       |
| titles |          1 | fk4      |            1 | emp_no      | A         |      284782 |     NULL |   NULL |      | BTREE      |         |              | YES     | NULL       |
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
3 rows in set (0.02 sec)
```

2.

```
mysql> DESCRIBE employees;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| emp_no      | int           | NO   | PRI | NULL    |       |
| birth_date  | date          | YES  |     | NULL    |       |
| first_name  | varchar(14)   | YES  | MUL | NULL    |       |
| last_name   | varchar(16)   | YES  |     | NULL    |       |
| sex         | enum('M','F') | YES  |     | NULL    |       |
| hire_date   | date          | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

```
mysql> SHOW INDEX FROM employees;
+-----------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
| Table     | Non_unique | Key_name   | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
| employees |          0 | PRIMARY    |            1 | emp_no      | A         |      299433 |     NULL |   NULL |      | BTREE      |         |              | YES     | NULL       |
| employees |          1 | fname_index |           1 | first_name  | A         |        1246 |     NULL |   NULL | YES  | BTREE      |         |              | YES     | NULL       |
+-----------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------+---------+------------+
2 rows in set (0.01 sec)
```

B.

Creat emplist table,

```
mysql> CREATE TABLE emplist SELECT emp_no, first_name FROM employees;
Query OK, 300024 rows affected (1.75 sec)
Records: 300024  Duplicates: 0  Warnings: 0
```

Creat titleperiod table,

```
mysql> CREATE TABLE titleperiod SELECT emp_no, title, DATEDIFF(to_date, from_date) AS period FROM titles;
Query OK, 421982 rows affected (4.83 sec)
Records: 421982  Duplicates: 0  Warnings: 0
```

1. Analyze the table,

```
mysql> DESCRIBE emplist;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| emp_no     | int         | NO   |     | NULL    |       |
| first_name | varchar(14) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

```
mysql> SHOW INDEX FROM emplist;
Empty set (0.00 sec)
```

2.

```
mysql> DESCRIBE titleperiod;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| emp_no | int         | NO   |     | NULL    |       |
| title  | varchar(50) | NO   |     | NULL    |       |
| period | int         | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

```
mysql> SHOW INDEX FROM titleperiod;
Empty set (0.00 sec)
```

C.

```
mysql> SELECT e.first_name, t.period
    -> FROM emplist e
    -> JOIN titleperiod t ON e.emp_no = t.emp_no
    -> WHERE t.period > 4000;
```

first_name	period
Aamer	2921719
Aamer	2922253
Aamer	2922706
Aamer	2922476
Aamer	2923169
Aamod	2922833
Aamod	2923478
Aamod	2922579
Aamod	2922133
Abdelaziz	2921865
Abdelaziz	2920732
Abdelaziz	2922727
Abdelaziz	2922274
Abdelaziz	2922985
Abdelaziz	2923224
Abdelaziz	2922907
Abdelaziz	2921978
Abdelaziz	2922138
Abdelaziz	2921620
Abdelaziz	2921719
Abdelaziz	2922152
Abdelghani	2922873
Abdelghani	2921677
Abdelwaheb	2921611
Abdelwaheb	2923522
Abdelwaheb	2923020
Abdelwaheb	2922736
Abdulah	2923264
Abdulah	2922155
Abdulah	2922249
Abdulah	2922419
Abdulla	2921348
Abdulla	2923822
Abdulla	2921944

D.Explain without index,

```
mysql> EXPLAIN SELECT e.first_name,t.period FROM emplist e
    -> JOIN titleperiod t ON e.emp_no = t.emp_no
    -> WHERE t.period >4000;
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
| id | select_type | table | partitions | type | possible_keys         | key        | key_len | ref                 | rows   | filtered | Extra       |
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
|  1 | SIMPLE      | e     | NULL       | ALL  | PRIMARY               | NULL       | NULL    | NULL                | 299212 |   100.00 | NULL        |
|  1 | SIMPLE      | t     | NULL       | ref  | idx_emp_no,idx_period | idx_emp_no | 4       | e21company.e.emp_no |      1 |    50.00 | Using where |
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
2 rows in set, 1 warning (0.04 sec)
```

- type: ALL - Full table scans on both tables
- rows: High number of rows examined
- Extra: Using where - filtering after reading all rows
- Performance: Poor due to nested loop joins without indexes

E.

Creat primary key on emplist,

```
mysql> ALTER TABLE emplist ADD PRIMARY KEY (emp_no);
Query OK, 0 rows affected (1.22 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Creat index on titleperiod,

```
mysql> CREATE INDEX idx_emp_no ON titleperiod(emp_no);
Query OK, 0 rows affected (0.90 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE INDEX idx_period ON titleperiod(period);
Query OK, 0 rows affected (0.87 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3.

```
mysql> EXPLAIN SELECT e.first_name,t.period FROM emplist e
    -> JOIN titleperiod t ON e.emp_no = t.emp_no
    -> WHERE t.period >4000;
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
| id | select_type | table | partitions | type | possible_keys         | key        | key_len | ref                 | rows   | filtered | Extra       |
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
|  1 | SIMPLE      | e     | NULL       | ALL  | PRIMARY               | NULL       | NULL    | NULL                | 299212 |   100.00 | NULL        |
|  1 | SIMPLE      | t     | NULL       | ref  | idx_emp_no,idx_period | idx_emp_no | 4       | e21company.e.emp_no |      1 |    50.00 | Using where |
+----+-------------+-------+------------+------+-----------------------+------------+---------+---------------------+--------+----------+-------------+
2 rows in set, 1 warning (0.00 sec)
```

Improved Analysis:

1. type: eq_ref/ref - Index-based joins
2. rows: Fewer rows examined
3. key: Shows which indexes are used
4. Performance: Significantly better

4.YES,

- Covering Index: Create composite index including all needed columns
- Query Rewriting: Consider different join orders
- Partitioning: For very large tables
- Statistics Update: Ensure MySQL has current table statistics

TASK 05

Query 1: Engineers per Department

```
mysql> CREATE INDEX idx__title ON titles(title);
Query OK, 0 rows affected (1.40 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE INDEX idx_dept_emp ON dept_emp(emp_no, dept_no);
Query OK, 0 rows affected (0.81 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_departments ON departments(dept_no);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW INDEX FROM titles;
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| titles |          0 | PRIMARY  |            1 | title       | A         |           5 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| titles |          0 | PRIMARY  |            2 | emp_no      | A         |      426987 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| titles |          1 | fk4      |            1 | emp_no      | A         |      284782 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| titles |          1 | idx__title |          1 | title       | A         |           5 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+--------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
4 rows in set (0.01 sec)
```

```
mysql> SHOW INDEX FROM dept_emp;
+----------+------------+----------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name             | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+----------+------------+----------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| dept_emp |          0 | PRIMARY              |            1 | emp_no      | A         |      299425 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          0 | PRIMARY              |            2 | dept_no     | A         |      331125 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | dept_no_to_dept_no   |            1 | dept_no     | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_dept_emp         |            1 | emp_no      | A         |      299773 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_dept_emp         |            2 | dept_no     | A         |      331125 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_current_dept     |            1 | emp_no      | A         |      300174 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_current_dept     |            2 | to_date     | A         |      331125 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_current_emp_dept |            1 | emp_no      | A         |      300149 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| dept_emp |          1 | idx_current_emp_dept |            2 | to_date     | A         |      331125 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+----------+------------+----------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
9 rows in set (0.04 sec)
```

```
mysql> EXPLAIN SELECT d.dept_name, COUNT(*) as num_engineers
    -> FROM departments d
    -> JOIN dept_emp de ON d.dept_no = de.dept_no
    -> JOIN titles t ON de.emp_no = t.emp_no
    -> WHERE t.title = 'Engineer'
    -> AND de.to_date = '9999-01-01'
    -> AND t.to_date = '9999-01-01'
    -> GROUP BY d.dept_no, d.dept_name;
+----+-------------+-------+------------+------+---------------------------------------------------------------------------------------------+---------------+---------+-------------------------+--------+----------+-----------------------------+
| id | select_type | table | partitions | type | possible_keys                                                                                | key           | key_len | ref                     | rows   | filtered | Extra                       |
+----+-------------+-------+------------+------+---------------------------------------------------------------------------------------------+---------------+---------+-------------------------+--------+----------+-----------------------------+
|  1 | SIMPLE      | t     | NULL       | ref  | PRIMARY,fk4,idx__title                                                                       | PRIMARY       | 202     | const                   | 196934 |    10.00 | Using where; Using temporary |
|  1 | SIMPLE      | de    | NULL       | ref  | PRIMARY,dept_no_to_dept_no,idx_dept_emp,idx_current_dept,idx_current_emp_dept | PRIMARY | 4 | e21company.t.emp_no | 1 |  10.00 | Using where                 |
|  1 | SIMPLE      | d     | NULL       | eq_ref | PRIMARY,idx_departments,idx_dept_name                                                       | PRIMARY       | 40      | e21company.de.dept_no   |      1 |   100.00 | NULL                        |
+----+-------------+-------+------------+------+---------------------------------------------------------------------------------------------+---------------+---------+-------------------------+--------+----------+-----------------------------+
3 rows in set, 1 warning (0.02 sec)
```

Expected Improvements:

- type: Changes from ALL to ref/eq_ref
- key: Shows which indexes are being used
- rows: Significant reduction in rows examined
- Extra: More efficient join algorithms

Query 3: High Salary Employees by Department

```
mysql> CREATE INDEX idx_salary ON salaries(emp_no, salary);
Query OK, 0 rows affected (4.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_current_dept ON dept_emp(emp_no, to_date);
Query OK, 0 rows affected (1.15 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW INDEX FROM salaries;
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| salaries |          0 | PRIMARY  |            1 | emp_no      | A         |      201167 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| salaries |          0 | PRIMARY  |            2 | from_date   | A         |     1873155 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| salaries |          0 | PRIMARY  |            3 | to_date     | A         |     1873155 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| salaries |          1 | idx_salary |          1 | emp_no      | A         |      196947 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| salaries |          1 | idx_salary |          2 | salary      | A         |     1873155 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
5 rows in set (0.03 sec)
```

```
mysql> EXPLAIN SELECT d.dept_name, t.title, COUNT(*) as employee_count
    -> FROM departments d
    -> JOIN dept_emp de ON d.dept_no = de.dept_no
    -> JOIN salaries s ON de.emp_no = s.emp_no
    -> JOIN titles t ON de.emp_no = t.emp_no
    -> WHERE s.salary > 115000
    -> AND s.to_date = '9999-01-01'
    -> AND de.to_date = '9999-01-01'
    -> AND t.to_date = '9999-01-01'
    -> GROUP BY d.dept_name, t.title;
+----+-------------+-------+------------+--------+---------------------------------------------------------------------------------------------+---------+---------+-------------------------+--------+----------+-------------------------------+
| id | select_type | table | partitions | type   | possible_keys                                                                                | key     | key_len | ref                     | rows   | filtered | Extra                         |
+----+-------------+-------+------------+--------+---------------------------------------------------------------------------------------------+---------+---------+-------------------------+--------+----------+-------------------------------+
|  1 | SIMPLE      | t     | NULL       | ALL    | fk4                                                                                          | NULL    | NULL    | NULL                    | 426987 |    10.00 | Using where; Using temporary |
|  1 | SIMPLE      | de    | NULL       | ref    | PRIMARY,dept_no_to_dept_no,idx_dept_emp,idx_current_dept,idx_current_emp_dept | PRIMARY | 4       | e21company.t.emp_no     |      1 |    10.00 | Using where                   |
|  1 | SIMPLE      | s     | NULL       | ref    | PRIMARY,idx_salary                                                                           | PRIMARY | 4       | e21company.t.emp_no     |      9 |     3.33 | Using where                   |
|  1 | SIMPLE      | d     | NULL       | eq_ref | PRIMARY,idx_departments                                                                      | PRIMARY | 40      | e21company.de.dept_no   |      1 |   100.00 | NULL                          |
+----+-------------+-------+------------+--------+---------------------------------------------------------------------------------------------+---------+---------+-------------------------+--------+----------+-------------------------------+
4 rows in set, 1 warning (0.00 sec)
```

Expected Improvements:

- Range scan on salary index instead of full table scan
- Efficient joins using emp_no indexes
- Reduced rows examined due to early filtering

Query 5: Non-HR Employees

```
mysql> CREATE INDEX idx_dept_name ON departments(dept_name);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_current_emp_dept ON dept_emp(emp_no, to_date);
Query OK, 0 rows affected, 1 warning (0.90 sec)
Records: 0  Duplicates: 0  Warnings: 1
```

TASK 06

- Negative Impacts:
    1. Slower INSERT: Each insert must update all relevant indexes
    2. Slower UPDATE: Updates to indexed columns require index maintenance
    3. Slower DELETE: Must remove entries from all indexes
    4. Storage Overhead: Indexes consume additional disk space

- Performance:
    1. Read-Heavy Workload: Indexes provide significant benefits
    2. Write-Heavy Workload: Indexes can degrade performance
    3. Mixed Workload: Balance between read and write performance